CLAUDINE CHILDCARE TRACKER

Final Project Report

Course: Mobile Application Development with Flutter

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Project: Childcare Tracker

Firebase Project: claudine-childcare-tracker

Lecturer: Dr. BUGINGO EMMANUEL

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1. Introduction

Purpose

Childcare Tracker is a Flutter mobile application designed to help a parent or authorized user organize information about children, daily activities, and developmental milestones in one place.

Overview

The application combines Flutter for the user interface, Firebase Authentication for account access, Cloud Firestore for structured application data, Cloud Storage for child images, and Google AdMob for advertising. The application uses an authentication gate so that protected application screens are available only after sign-in.

Scope

The project implements four interrelated entities required by the assignment: User, Child, Activity, and Milestone. Children belong to a parent user, while activities and milestones are linked to both a child and the parent user.

2. Problem Statement

Problem

Parents and caregivers may keep child information in separate notes, messages, or paper records. This makes it difficult to organize child profiles, daily activities, and developmental progress consistently.

Proposed solution

Childcare Tracker provides a single application where an authenticated user can create and manage child records, record activities, and record milestones. Firebase provides cloud-based persistence and access control for the application data.

3. Objectives

General objective

To develop a Flutter-based childcare tracking application that securely stores and manages child-related information using Firebase services.

Specific objectives

1) Implement registration, sign-in, sign-out, and an authentication gate. 2) Implement CRUD operations for User, Child, Activity, and Milestone data. 3) Connect activities and milestones to the correct child and parent. 4) Support child image handling through Cloud Storage. 5) Add banner and interstitial AdMob test advertisements without allowing ad failure to break the application. 6) Apply Firestore and Storage security rules based on authenticated ownership.

4. Tools and Technologies

Technologies

Flutter and Dart were used for application development. Firebase Authentication handles user authentication. Cloud Firestore stores application entities. Cloud Storage is prepared for image storage. Google Mobile Ads/AdMob provides test advertisements. Git and GitHub are used for version control and project submission. Visual Studio Code and the Flutter command-line tools are used during development.

Project structure

The project contains separate folders for screens, services, widgets, Firebase configuration, rules, and supporting Flutter files. A screenshots folder is also included for project evidence.

5. Application Design

Entity relationships

User → Child: a parent user owns child records through parentId. Child → Activity: an activity contains childId and parentId. Child → Milestone: a milestone contains childId and parentId. These relationships allow the application to show information for the correct authenticated user and selected child.

Main screens

The application includes authentication screens, a home/children screen, an activities screen, and a milestones screen. Navigation is provided from the home screen drawer.

Data fields

User includes userId, fullName, email, phone, role, and profileImageUrl. Child includes parentId, name, gender, dateOfBirth, photoUrl, and notes. Activity includes childId, parentId, activityType, description, activityDate, durationMinutes, and notes. Milestone includes childId, parentId, title, description, milestoneDate, category, achieved, and photoUrl.

6. Implementation — Milestone Log

Milestone	Work completed
M1 — Project setup	Created the Flutter project and organized the application into screens, services, widgets, and Firebase configuration.
M2 — Firebase connection	Connected the application to the Firebase project claudine-childcare-tracker and configured Firebase initialization.
M3 — Authentication	Implemented sign-up, sign-in, sign-out, and an authentication gate using Firebase Authentication.
M4 — Firestore entities	Implemented User, Child, Activity, and Milestone collections with the required relationships.
M5 — CRUD operations	Implemented creation, reading, updating, and deletion flows for child, activity, and milestone records.
M6 — Images and Storage	Added image-picking and image-display code and prepared Cloud Storage upload and security-rule support. Storage activation remained unavailable in the Firebase console during development because the project required an upgrade.
M7 — AdMob	Added a banner ad widget and an interstitial ad service using Google test ad unit IDs. Ad initialization and failures are handled so advertising does not stop the application.
M8 — Testing and submission preparation	Ran flutter analyze successfully with no remaining analysis issues, tested Firestore persistence and navigation, organized screenshots, and pushed the project to GitHub.

Screenshot evidence: The project screenshots are stored in the repository folder named *screenshoot*. Insert or place the relevant screenshots beside the corresponding milestone in the final submitted report if the lecturer requires screenshots inside the PDF.

7. Testing

Area	Test	Result
Authentication	Register/sign in/sign out	Successful authentication flow observed.
Auth Gate	Unauthenticated access	User is directed to authentication screens.
Child CRUD	Create/read/update/delete child	Child data persisted in Firestore.
Activity CRUD	Create/read/update/delete activity	Activity data linked to selected child.
Milestone CRUD	Create/read/update/delete milestone	Milestone data linked to selected child.
Ownership	parentId filtering	Records are queried using the authenticated user ID.
Persistence	Refresh/reopen data	Firestore records remain available after refresh.
Validation	Required form fields	Invalid/incomplete input is prevented or reported.
Ad failure safety	Ad loading failure	Ad failure is caught so the main app can continue.
Code quality	flutter analyze	No remaining analysis issues were reported during final checking.

8. Challenges Encountered

During development, several technical challenges were encountered. Android emulator setup was not completed on the development computer, so Chrome/web was used for much of the functional development and testing. AdMob test ads are intended for supported mobile platforms, so their complete live demonstration requires an Android device or emulator. Firebase Cloud Storage was also unavailable because the Firebase console required an upgrade before Storage could be activated. The Storage integration and rules were therefore prepared in the project, but actual cloud image upload could not be fully demonstrated under the available Firebase project configuration.

Another challenge was maintaining secure ownership relationships. The Firestore rules were written so that authenticated users can access records associated with their own user ID. The application also uses parentId and childId fields to connect the four entities.

9. Limitations

The main limitation at the time of report preparation is Cloud Storage activation. Because the Firebase console required an upgrade, actual Storage upload testing could not be completed. The application retains the Storage service and rules so the feature can be activated and tested when Storage is available.

A second limitation is that full AdMob demonstration depends on a supported Android environment with internet access. The project uses Google test ad units and includes failure handling, but web development cannot be treated as the final Android advertising demonstration.

10. Conclusion

Childcare Tracker demonstrates the integration of Flutter with Firebase Authentication and Cloud Firestore, together with prepared Cloud Storage and AdMob functionality. The application provides four related entities, authenticated access, CRUD operations, ownership-based data relationships, navigation, validation, and advertising support. The project was analyzed successfully and its source code and screenshots were organized for submission. Remaining limitations are documented rather than hidden, particularly the Firebase Storage activation and Android advertising demonstration requirements.

References and Submission Information

1. Flutter documentation and Dart language documentation used as development references.

2. Firebase documentation for Authentication, Cloud Firestore, Cloud Storage, and security rules.

3. Google Mobile Ads / AdMob documentation for Flutter mobile advertising and test ad units.

4. Course practical assignment: Mobile Application Development with Flutter, Weekend/Practical Project Days 10–12, Dr. BUGINGO EMMANUEL.

GitHub Repository

https://github.com/cdine250/childcare_tracker

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Screenshot note: Actual project screenshots are stored separately in the repository folder named “screenshoot”. They should be inserted into this PDF where required by the lecturer before final submission.